

Machine Learning I: Hands-On 9 - Worked Solutions

9.1 From ordinary datasets to meta-features: Worked Solution

The original table describes five complete classification datasets:

Dataset	Class 0 examples	Class 1 examples	Raw features	Numeric features	Missing values	Total values
D1	10	10	2	2	0	40
D2	18	2	3	3	0	60
D3	6	4	10	10	5	100
D4	14	6	5	2	4	100
D5	8	8	4	3	8	64

The objective is to convert these dataset-level summaries into meta-features. Each row of the final table will describe an **entire dataset**, not an individual observation.

1. Basic dataset-level meta-features

For the number of instances, add the two class counts. For the ratios, use

$$\text{Numeric Ratio} = \frac{\text{Numeric features}}{\text{Raw features}}, \quad \text{Missing Rate} = \frac{\text{Missing values}}{\text{Total values}}.$$

Dataset	Number of instances	Number of raw features	Numeric feature ratio	Missing value rate
D1	$10 + 10 = 20$	2	$2/2 = 1.00$	$0/40 = 0.000$
D2	$18 + 2 = 20$	3	$3/3 = 1.00$	$0/60 = 0.000$
D3	$6 + 4 = 10$	10	$10/10 = 1.00$	$5/100 = 0.050$
D4	$14 + 6 = 20$	5	$2/5 = 0.400$	$4/100 = 0.040$
D5	$8 + 8 = 16$	4	$3/4 = 0.750$	$8/64 = 0.125$

2. Class-distribution meta-features

For each dataset,

$$\text{Majority proportion} = \frac{\max(n_0, n_1)}{n_0 + n_1},$$

$$\text{Minority proportion} = \frac{\min(n_0, n_1)}{n_0 + n_1},$$

and

$$\text{Imbalance ratio} = \frac{\text{Majority proportion}}{\text{Minority proportion}}.$$

Dataset	Majority class proportion	Minority class proportion	Imbalance ratio
D1	$10/20 = 0.50$	$10/20 = 0.50$	$0.50/0.50 = 1.00$

Dataset	Majority class proportion	Minority class proportion	Imbalance ratio
D2	$18/20 = 0.90$	$2/20 = 0.10$	$0.90/0.10 = 9.00$
D3	$6/10 = 0.60$	$4/10 = 0.40$	$0.60/0.40 = 1.50$
D4	$14/20 = 0.70$	$6/20 = 0.30$	$0.70/0.30 \approx 2.33$
D5	$8/16 = 0.50$	$8/16 = 0.50$	$0.50/0.50 = 1.00$

3. Final meta-dataset

Combining the previous calculations gives:

Dataset	Instances	Raw features	Numeric ratio	Missing rate	Majority prop.	Minority prop.	Imbalance ratio
D1	20	2	1.00	0.000	0.50	0.50	1.00
D2	20	3	1.00	0.000	0.90	0.10	9.00
D3	10	10	1.00	0.050	0.60	0.40	1.50
D4	20	5	0.400	0.040	0.70	0.30	2.33
D5	16	4	0.750	0.125	0.50	0.50	1.00

4. Why is this a meta-dataset?

This is a meta-dataset because each row represents one complete dataset: D1, D2, D3, D4, or D5. The columns are not ordinary predictor variables measured for individual examples. They are **meta-features** that summarize properties of complete learning problems, such as sample size, dimensionality, missingness, and class imbalance.

In an ordinary classification dataset, a row might describe one patient, image, transaction, or other instance. Here, a row describes an entire classification problem.

5. Which datasets might be more difficult for a simple classifier?

Based only on the supplied meta-features, D2 and D3 are reasonable candidates.

- **D2:** strong class imbalance. The majority-class proportion is 0.90, the minority-class proportion is 0.10, and the imbalance ratio is 9.00. A classifier can therefore achieve apparently high accuracy while performing poorly on the minority class.
- **D3:** high dimensionality relative to its small sample size, together with missing values. It has 10 instances and 10 raw features, giving a feature-to-instance ratio of 1, and its missing-value rate is 0.05.

ML insight: meta-features are evidence, not performance

A meta-feature describes properties of a task. It can help explain or predict algorithm behaviour across tasks, but it does not directly measure how well a particular model will generalize on that task.

9.2 Meta-learning for algorithm selection: Worked Solution

A full decision tree and a decision stump were evaluated on six datasets:

Dataset	Complexity	Imbalance	Missingness	Accuracy_DT	Accuracy_DS
D1	0.10	0.00	0.00	0.90	0.80
D2	0.15	0.90	0.00	0.80	0.90
D3	0.90	0.20	0.10	0.60	0.70
D4	0.10	0.00	0.00	0.95	0.65
D5	0.30	0.20	0.05	0.85	0.75

Dataset	Complexity	Imbalance	Missingness	Accuracy_DT	Accuracy_DS
D6	0.25	0.00	0.15	0.75	0.75

1-2. Accuracy gaps and best model

For every dataset,

$$\text{Accuracy_Gap} = \text{Accuracy_DT} - \text{Accuracy_DS}.$$

Dataset	Accuracy_DT	Accuracy_DS	Calculation	Gap	Best model
D1	0.90	0.80	$0.90 - 0.80$	0.10	Full decision tree
D2	0.80	0.90	$0.80 - 0.90$	-0.10	Decision stump
D3	0.60	0.70	$0.60 - 0.70$	-0.10	Decision stump
D4	0.95	0.65	$0.95 - 0.65$	0.30	Full decision tree
D5	0.85	0.75	$0.85 - 0.75$	0.10	Full decision tree
D6	0.75	0.75	$0.75 - 0.75$	0.00	Tie

The sign of the gap tells us which learner performed better on each previous task. Those observed outcomes become the **meta-level labels** used for algorithm selection.

3. Distances from the query dataset

The query is

$$Q = (0.12, 0.05, 0.00).$$

Using Manhattan distance,

$$d(Q, D_i) = |C_Q - C_i| + |I_Q - I_i| + |M_Q - M_i|.$$

The calculations are:

Dataset	Meta-features	Calculation	Distance
D1	(0.10, 0.00, 0.00)	$ 0.12 - 0.10 + 0.05 - 0.00 + 0.00 - 0.00 = 0.02 + 0.05 + 0.00$	0.07
D2	(0.15, 0.90, 0.00)	$ 0.12 - 0.15 + 0.05 - 0.90 + 0.00 - 0.00 = 0.03 + 0.85 + 0.00$	0.88
D3	(0.90, 0.20, 0.10)	$ 0.12 - 0.90 + 0.05 - 0.20 + 0.00 - 0.10 = 0.78 + 0.15 + 0.10$	1.03
D4	(0.10, 0.00, 0.00)	$ 0.12 - 0.10 + 0.05 - 0.00 + 0.00 - 0.00 = 0.02 + 0.05 + 0.00$	0.07
D5	(0.30, 0.20, 0.05)	$ 0.12 - 0.30 + 0.05 - 0.20 + 0.00 - 0.05 = 0.18 + 0.15 + 0.05$	0.38
D6	(0.25, 0.00, 0.15)	$ 0.12 - 0.25 + 0.05 - 0.00 + 0.00 - 0.15 = 0.13 + 0.05 + 0.15$	0.33

Sorting from closest to farthest gives:

Order	Dataset	Distance	Best model on that dataset
1	D1	0.07	Full decision tree
1	D4	0.07	Full decision tree
3	D6	0.33	Tie
4	D5	0.38	Full decision tree
5	D2	0.88	Decision stump
6	D3	1.03	Decision stump

D1 and D4 are tied as the closest datasets because they have identical meta-features (0.10, 0.00, 0.00).

4. 1-nearest-neighbor recommendation

The nearest distance is 0.07, attained by both D1 and D4. Both have the same meta-level label: **Full decision tree**.

Therefore, even though there is a distance tie, the recommendation is unambiguous:

Recommend the full decision tree.

5. 3-nearest-neighbor recommendation

The three closest datasets are:

Neighbor	Distance to Q	Best model
D1	0.07	Full decision tree
D4	0.07	Full decision tree
D6	0.33	Tie

The vote is therefore:

- Full decision tree: 2 votes;
- Decision stump: 0 votes;
- Tie / stump-or-tie: 1 vote.

The majority vote selects

Full decision tree.

6. Why is this meta-learning?

This is a meta-learning problem because the examples are **datasets**, not ordinary observations.

- **Examples:** D1, D2, D3, D4, D5, and D6.
- **Predictors:** dataset-level meta-features such as Complexity, Imbalance, and Missingness.
- **Target:** the algorithm recommendation for the dataset - full decision tree, decision stump, or tie.

The goal is therefore is to use experience from previous learning tasks to recommend a learning strategy for a new task.

ML insight: algorithm selection is itself a learning problem

The recommendation is only as reliable as the previous tasks used to construct the meta-dataset. A small or unrepresentative collection of datasets can produce a poor recommendation even when the meta-level learner is applied correctly.